



MILL TEST CERTIFICATE

Customer: DNOW L.P. PO BOX 40985 Houston,TX,77240-0985 281-823-4700	Sales Order: 0630004052 PO: 26502109	Invoice No.: 1600045043 Date of Issue: 03/09/2023
Certificate No.: 1600045043-2		
Product Description: 5L 12.750 0.250 33.41# X52BDRH5BW-20P2		
Specification: 5L 46th Edition May 2019		

AXIS PIPE AND TUBE, LLC.
1451 Louis E. Mikulin Rd.
Bryan TX 77807
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Heat Number		Lot Number		Mat Number		Steel Making / Coil Rolling																					
B215712		LT00018684		333369		Steel Dynamics Columbus, LLC , Columbus - MS - US / Steel Dynamics Columbus, LLC , Columbus - MS - US																					
Mechanical Properties																											
Tensile Test - Strip Specimen Gauge Length 2"								Temp (F) -20				CVN				Temp (F) -20						Size	Loc. / Orient	Acceptance Criteria (ft-lbf)			
Loc. / Orient	Size	Y.S (psi)	U.T.S. Body (psi)	U.T.S. Weld (psi)	Y/T Ratio	E.L. (%)	Hardness (HRB)	Body (ft-lbf)				Shear Area (%)				Weld (ft-lbf)				Shear Area (%)							
								1	2	3	Avg	1	2	3	Avg	1	2	3	Avg	1	2	3	Avg				
T180°	1 1/2"	64500	74400	72800	.87	34.7	83	88	80	82	84	90	90	90	90	91	105	88	95	90	95	90	92	10x5.0 mm	3T90	8	
								Lateral Expansion (in)				.071	.068	.064	0.068	Lateral Expansion (in)				.068	.063	UB	0.066				

Analysis (H/P)	Chemical Composition (Max %)																	
	C	Si	Mn	P	S	Ni	Cr	Mo	Cu	V	Al	Nb	Ti	B	Ca	N	CEpcm (API Max. A.C.)	CE Results
H	0.06	0.230	0.85	0.009	0.002	0.05	0.09	0.01	0.14	0.002	0.020	0.026	0.016	0.0003	0.0019	0.0064	0.25	0.13
P	0.06	0.230	0.83	0.012	0.004	0.04	0.06	0.01	0.13	0.003	0.019	0.023	0.013	0.0005	0.0019	0.0083	0.25	0.12
P	0.05	0.230	0.82	0.011	0.002	0.05	0.09	0.01	0.14	0.003	0.019	0.020	0.013	0.0003	0.0020	0.0061	0.25	0.11

Following Controls / Tests have been satisfactorily performed:		
1.- Hydrostatic Testing:	2,050.00 psi Pressure for	5.00 seconds
2: Ultra Sonic Tested to N10 - passed		
3: Seam Annealed min 1,600 F		
4: Flattening Test Passed		

Acronyms:	
L = Longitudinal	T = Transverse
Y.S. = Yield Strength	U.T.S. = Ultimate Tensile Strength
E.L. = Elongation	H = Heat Analysis
P = Product Analysis	A.C. = Acceptance Criteria
UB = Un-Broken	N/A = Not Applicable

Remarks:
1: This material meets:Grades ASTM A53/A53M-22 Grade B API 5L HFW PSL2 X52M,
2: Meets Hardness Requirements of NACE MRO 175 / ISO 15156-2 clause A.2.1.1
3: Made and Melted in USA
4: Date of Manufacture 03/2023
5: This product is silicon and/or aluminum fully killed.

We hereby certify that the material herein has been made and tested in accordance with the above specification and also with the requirement called for by the above order.

Quality Manager